results. These effects can also be previewed in viewports. The downside of using multi-pass effects is that they can slow rendering significantly, because each pass requires that each frame be completely rendered (so 12 passes will render 12 times as slow).

The Depth of Field render effect is applied by choosing Rendering > Effects and adding the

Depth of Field effect to the Effects list. Render effects are post-processing effects, so the scene is rendered by the scanline renderer before the effect is applied. Depth of Field works by calculating the depth of each pixel in the rendered scene, and then dividing the scene into foreground, background, and in-focus pixels. The foreground and background images are then blurred, and the final image is composited from the processed originals.

Depth of field in mental ray is much more precise. It is similar to how depth of field is calculated in a real camera, in that mental ray uses an f-stop parameter to control the amount of blurring. Setting up depth of field is similar to the multi-pass effect; the camera's target distance determines the focal plane of the camera. In mental ray, motion blur is configured by selecting the Depth of Field (mental ray) option in the Multi-Pass effect. When this is selected, the Depth of Field parameters give you one option to set the f-stop. The f-stop parameter controls the amount of blurring at distances away from the focal plane. In a real-world camera, the f-stop measures the size of the lens's aperture. The lower the f-stop value, the larger the aperture and more the blurring. Mental ray works the same: lower numbers enhance the effect; higher numbers attenuate it.

5.3.5 Motion Blur

Another camera effect is motion blur. In the real world, it takes time to expose a frame of film. If the subject moves during this exposure time, the resulting image is blurred. In 3ds Max, there is no such thing as film speed, so motion blur has to be simulated. This is done in several ways, depending on the renderer. 3ds Max's scanline renderer has three types of motion blur: Image, Object, and Multi-Pass.

Effect of motion blur while rendering